

Internet Archive

Proposal - Google Summer of Code 2026

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Project Proposal

Project Title

Scalable Tag Normalization and Classification Pipeline for Open Library

Overview

Open Library contains millions of book records enriched with subject tags describing themes, genres, places, and other attributes. While the platform already provides an end-to-end subject pipeline—including Work metadata fields, bulk editing tools, Solr-based indexing, and subject discovery pages—these tags remain inconsistent, duplicated, and loosely structured.

This project proposes a safe, deterministic pipeline to analyze, normalize, and migrate subject metadata at scale. The system will integrate directly with Open Library's existing architecture and focus on improving metadata quality without disrupting current workflows.

The implementation will begin with a high-impact, scoped target (e.g., genre normalization within existing subject structures) and will validate improvements through Solr facet consistency and discoverability metrics.

Problem Statement

Subject tags in Open Library are contributed from diverse sources and stored as semi-structured strings across Work records. As a result:

- Duplicate and synonymous tags exist (e.g., "romance", "love story", "romantic fiction")
- Tags are inconsistently formatted (case, plurality, spelling variations)
- Tags are overloaded and lack clear categorization
- Search facets and subject browsing are less reliable due to noisy metadata

Although Open Library distinguishes subject types (subjects, subject_people, subject_places, subject_times), these are not consistently normalized.

This leads to reduced discoverability and inconsistent browsing experience.

The core challenge is to build a system that can:

- Normalize and deduplicate subject tags at scale
- Map tags to canonical forms using controlled vocabularies
- Safely update large volumes of Work records
- Maintain compatibility with Solr indexing and subject discovery systems

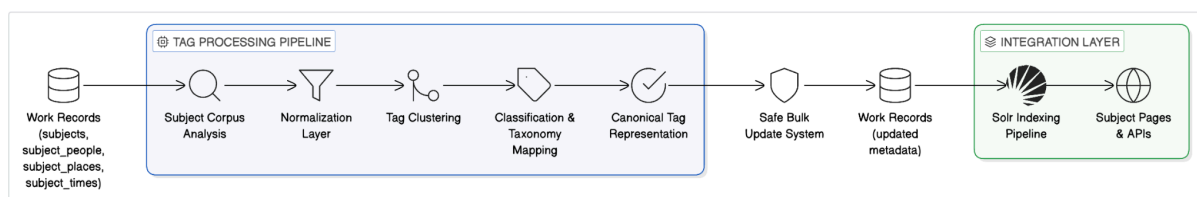
System Understanding

Open Library already provides a comprehensive subject pipeline:

- Subject data is stored on Work records (subjects, subject_people, subject_places, subject_times)
- Bulk updates are supported via existing tools (e.g., bulk_tag functionality)

- Subject data is indexed into Solr using WorkSolrBuilder, enabling faceted search and discovery
- Subject pages and APIs are powered by subject-based queries and facet aggregation

This project builds on top of this existing system rather than replacing it, focusing on improving metadata quality while preserving system stability and compatibility.



Relevant Code Areas

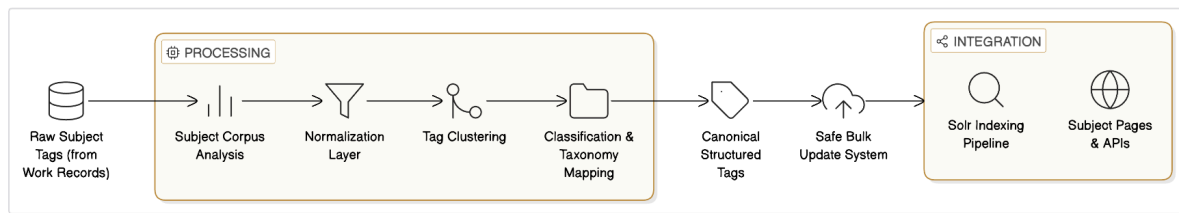
The implementation will interact with and extend the following components:

- `openlibrary/catalog/marc/get_subjects.py` (subject extraction patterns)
- `openlibrary/plugins/openlibrary/bulk_tag.py` (bulk updates)
- `openlibrary/solr/updater/work.py` (Solr indexing pipeline)
- `openlibrary/plugins/worksearch/subjects.py` (subject discovery)
- Solr subject indexing scripts (`index_subjects.py`)

Proposed Solution

Core Approach

I propose a modular pipeline that performs subject corpus analysis, normalization, clustering, classification, and safe migration, integrated with Open Library's existing metadata and indexing systems.



1. Subject Corpus Analysis

The pipeline begins by extracting subject data from Work records (**subjects**, **subject_people**, **subject_places**, **subject_times**).

Using Solr facet queries and indexing data, the system will identify:

- high-frequency tags
- duplicate and near-duplicate tags
- ambiguous or overloaded tags

This step produces a corpus analysis report that guides normalization and mapping decisions.

2. Normalization Layer

Subject tags are standardized using deterministic rules:

- case normalization
- duplicate removal
- canonical formatting
- handling common spelling variations

This ensures consistent representation of tags before further processing.

3.Tag Clustering

In this stage, semantically similar tags are grouped using deterministic and rule-based methods, such as:

- string similarity (e.g., edit distance)

- curated synonym mappings
- predefined normalization rules

This step avoids aggressive or probabilistic transformations and focuses on safe grouping of clearly equivalent tags.

Example:

"Romantic Fiction", "love story", "romance" → grouped into → "romance"

4. Classification & Taxonomy Mapping

Clustered tags are mapped into structured categories aligned with Open Library's existing subject model:

- subject
- subject_people
- subject_places
- subject_times

The system will initially focus on a high-impact category such as genre normalization, with extensions to other categories as future work.

5. Canonical Structured Tags

After classification, tags are transformed into a consistent canonical format:

- one standard representation per concept
- aligned with controlled vocabulary
- reusable across all records

This ensures that identical concepts are represented uniformly across the system.

6. Safe Bulk Update Pipeline

Canonical tags are applied back to Work records through a safe batch update mechanism.

This system:

- processes records in batches
- supports dry-run validation
- logs all transformations
- preserves original data where necessary

It leverages existing infrastructure (e.g., `bulk_tag`) to ensure safe large-scale updates.

7. Integration with Search and Discovery

After updates, Work records are re-indexed through the Solr indexing pipeline.

The impact of normalization is evaluated through:

- improved subject facet consistency
- reduced duplication in search filters
- better browsing experience on subject pages and APIs

Scope and Limitations

- Some tags may remain ambiguous and require manual review
- Perfect semantic classification may not always be achievable
- Introduction of new categories (e.g., genre) will be handled conservatively and may be scoped as an extension

The system will prioritize correctness, safety, and incremental improvements over aggressive transformations.

Deliverables

- Subject corpus analysis tools using Solr facet data
- Tag normalization and clustering pipeline
- Controlled taxonomy specification
- Safe bulk update system for subject metadata
- Integration with Solr indexing and subject discovery systems
- Evaluation reports and validation metrics
- Documentation

Evaluation

The system will be evaluated based on:

- Reduction in duplicate and inconsistent subject tags
- Increase in normalized and canonicalized tags
- Coverage of classified subject data
- Improvements in search facet quality and discoverability

Baseline:

Current unstructured subject metadata

Success Criteria:

- Significant reduction in redundant tags

- Improved consistency across records
- Observable improvement in search and browsing experience

Testing Strategy

- Unit tests for normalization and clustering logic
- Integration tests on real Open Library data samples
- Validation of Solr indexing and facet behavior
- Edge case handling:
 - ambiguous tags
 - rare or unique tags
 - large-scale batch processing

Timeline (12 Weeks)

Community Bonding

- Study Open Library architecture and subject pipeline
- Analyze existing subject data
- Finalize taxonomy and system design

Weeks 1–3

- Implement subject analysis tools
- Build normalization pipeline
- Run initial experiments on sample datasets

Weeks 4–6

- Implement clustering and classification system
- Define controlled vocabulary
- Evaluate normalization accuracy

Weeks 7–8

- Develop batch processing and safe update system
- Implement dry-run and logging mechanisms

Weeks 9–10

- Integrate with Solr indexing and subject discovery
- Validate search and facet improvements

Weeks 11–12

- Testing and performance tuning
- Documentation
- Final validation and reporting

Risk Mitigation

- Data loss risk → mitigated through dry-run mode, logging, and incremental updates
- Ambiguous tags → handled with conservative mapping and optional review
- Large-scale processing challenges → addressed through batching and controlled execution
- Search/index regressions → validated through before/after comparisons

About Me

I have experience in backend development and system design.

I have contributed to Open Library with an active pull request ([PR #11615](#)), where I refactored and unified the book provider download options system by replacing multiple provider-specific templates with a single structured rendering pipeline. This involved working with data models, templates, and ensuring compatibility across multiple providers.

This experience has given me familiarity with Open Library's architecture, including data modeling and rendering systems, motivating me to work on improving metadata quality at scale.

I am particularly interested in building scalable data processing systems that improve discoverability and usability.

Final Statement

This project focuses on improving metadata quality in Open Library by transforming unstructured subject tags into structured, standardized information. By building a scalable and reliable processing pipeline integrated with existing systems, it aims to significantly enhance discoverability, search quality, and user experience while maintaining system stability and data integrity.